Prevalence and Priorities: Characterizing Challenging Behaviors across Caregiver and Clinician Report Measures to Inform Intervention Targets in Children with Syndromic IDD

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WINGS
WEB INTERVENTION FOR PARENTS OF YOUTH WITH GENETIC SYNDROMES

RUSH

Background

- Behavior rating scales are part of best practices for evaluating outcomes in RCTs. However, these measures may not capture behaviors specific to an individual or caregiver treatment priorities.[1]
- **Web INtervention for youth with Genetic Syndromes** (WINGS) is a behavioral intervention RCT designed for syndromic IDD populations that utilizes both behavior rating scales and patient-reported outcome measures to inform intervention target behaviors.
- We used a **multimethod approach**, including the behavior checklist, **Behavior Problems Inventory (BPI-01)**, with the **Parent Top Problems (PTP)**, a family-centered, open-ended interview, to develop representative outcomes for trials in autism and IDD populations.[2,3]

Objectives

1. Characterize **challenging behaviors** that inform intervention goals in the cohort using both the **PTP**, a caregiver-centered measure, and the **BPI-01**, a behavior rating scale.
2. Identify **similarities and differences between the PTP** (patient-centered outcome) and **BPI-01** (severity/frequency scale) in characterizing challenging behavior profiles.

Participants

37 caregivers of eligible participants with syndromic IDD completed the BPI-01 during clinical screening and the PTP with an Independent Evaluator (IE) during baseline according to established protocols (See Table 1 for Measure Overview).

Child Demographics

- ***Age***: 2-12 years, *M* = 6.68, *SD* = 3.25
- ***Sex assigned at birth***: Male 75.68%, *n* = 28; Female 24.32%, *n* = 9
- ***Co-occurring autism***: 48.68%, *n* = 18
- ***Genetic Syndrome***
  - *Angelman Syndrome:* 16.22%, *n* = 6
  - *Fragile X Syndrome:* 21.62%, *n* = 8
  - *Phelan-McDermid Syndrome:* 29.73%, *n* = 11
  - *Other:* 32.43%, *n* = 11
- ***ID Level***
  - *Borderline ID:* 5.41%, *n* = 2
  - *Mild ID:* 45.95%, *n* = 17
  - *Moderate ID:* 37.84%, *n* = 14
  - *Severe ID:* 10.81%, *n* = 4

Table 1. PTP and BPI-01 Measure Overview

	PTP	BPI-01
Description	**Semi-structured clinician elicitation** of parent target problems.[2]	52-item **respondent-based behavior rating measure** across three subscales: 1) self-injurious, 2) stereotyped, and 3) aggressive/destructive behavior.[3]
Outcome	Each target problem is given a **label** and **narrative** describing the topography, frequency, context, and impact of the behavior.[2]	Each behavior is rated on **frequency** (*never, monthly, weekly, daily, hourly*) and **severity** (*mild, moderate, severe*), yielding a frequency and severity score for all three subscales.[3]
Behavior Examples	• ***Aggression***: Aggression, face grabbing/scratching, biting • ***Compliance Issues***: Noncompliance, difficulty with/resisting transitions • ***Safety Concerns***: Elopement, awareness/safety	• ***Stereotyped Behavior***: Repetitive body movements, manipulating objects • ***Self-Injurious Behavior***: Hitting head, teeth grinding, self-biting • ***Aggressive/Destructive Behavior***: biting, scratching, grabbing/pulling[3]

Figure 1. PTPs: Caregiver Reports of High-Priority Challenging Behaviors for Intervention Targets

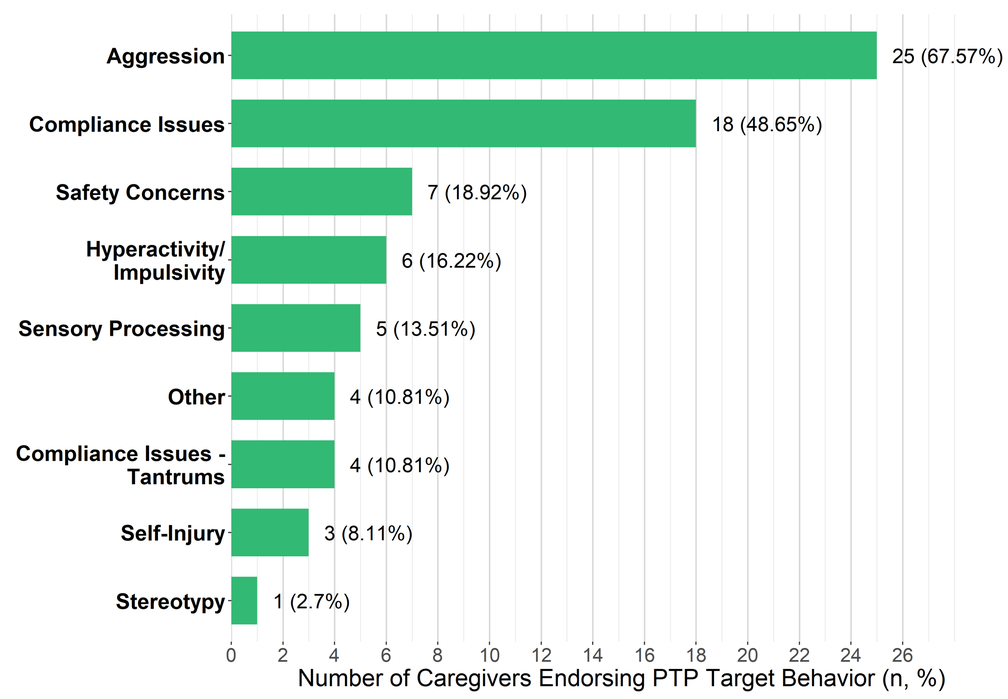
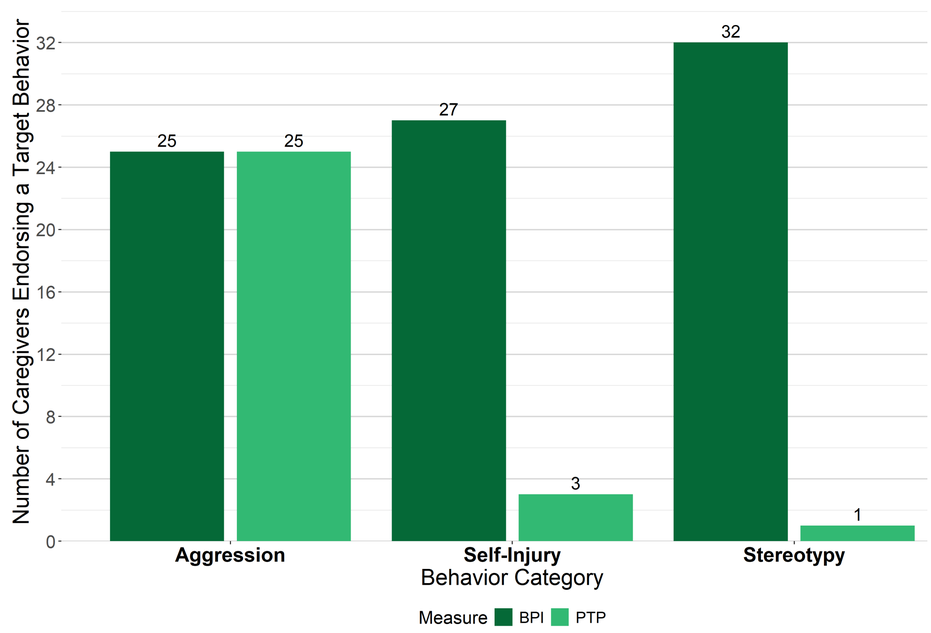


Table 2. BPI-01: Caregivers Endorsing Challenging Behaviors by Frequency and Severity Criteria

BPI Subscale	≥1 behavior on a Daily/Hourly Basis	≥1 behavior on a Daily/Hourly Basis also rated as Moderate or Severe
	Caregivers *n* (%)	
Aggressive/ Destructive Behavior	25 (67.57%)	25 (67.57%)
Self-Injurious Behavior	27 (72.97%)	24 (64.86%)
Stereotyped Behavior	32 (86.49%)	29 (78.38%)

Figure 2. Shared Topographies Across the PTP and BPI-01



Procedure

- **76 PTP labels** were categorized into 9 topography categories using methodology in previous RCTs[1] (Figure 1).
- Initial agreement on PTP categorization was 75% and reviewers reviewed the remainder to reach 100% agreement.
- We examined endorsement of Aggression, Stereotypies, and Self-Injury using the following criteria:
  - Endorsement of **at least 1 target problem on the PTP**
  - Endorsement of **at least 1 behavior which occurs on Daily or Hourly basis** on the **BPI-01**[4] (Table 2, Column 3)

Results

- Patient-reported outcomes: The **top three** most frequently endorsed challenging behaviors on the **PTP** were **Aggression** (67.57%, *n* = 25), **Compliance Issues** (48.65%, *n* = 18), and **Safety Concerns** (18.92%, *n* = 7) (Figure 1).
- Rating scale: When examining caregivers who endorsed **≥1 behavior on a Daily/Hourly basis on the BPI-01**, 67.57% caregivers (*n* = 25) endorsed Aggressive/Destructive, 72.97% (*n* = 27) endorsed Self-Injurious, and 86.49% (*n* = 32) endorsed Stereotyped behaviors (Table 2, Column 3).
- Rating scale: When examining caregivers who endorsed **≥1 behavior on a Daily/Hourly basis also rated Moderate or Severe on the BPI-01**, endorsement on the Aggressive/Destructive subscale was the same, but slightly lower for the Self-Injurious and Stereotyped Behavior subscales (Table 2, Column 3).
- Shared topographies: 67.57% (*n* = 25) caregivers endorsed **Aggression** on the PTP and BPI-01, while patterns across **Self-Injury** and **Stereotypy differed** (Figure 2).

Discussion

- While certain topographies, such as **Stereotypies** and **Self-Injury**, may be **highly prevalent** amongst autism and IDD populations, our data suggest that these **are not treatment priorities for caregivers**.
- Differences in the top behavior topography endorsed on the PTP (Aggression) and BPI-01 (Stereotypy) in our cohort demonstrate **the value of multimethod approaches** that include open-ended measurement along with standardized measurement tools to develop **meaningful, patient-centered treatment outcomes.**

WINGS Website

We would like to thank all of the families who made this research possible, the team at Emory University for their collaboration and expertise, and the NIH National Institute of Child Health and Human Development (NICHD) for funding this study (1R01HD107522).

ClinicalTrials.gov NCT06139172

References